Linear Algebra

Part – A

Fields: $\mathbb{F}$

Suppose we have a set $\mathbb{F}$ and two operations $+$ and $\cdot$

(I) addition: $x, y \in \mathbb{F}$ $\quad x+y \in \mathbb{F}$

(II) multiplication: $x, y \in \mathbb{F}$ $\quad xy \in \mathbb{F}$

$(\mathbb{F}, +, \cdot)$ is a field if it satisfies the following:

(i) Addition is commutative $\quad x+y = y+x \quad \forall\, x, y \in \mathbb{F}$

(II) Addition is associative $\quad x+(y+z) = (x+y)+z \quad \forall\, x, y, z \in \mathbb{F}$

(III) $\exists$ a unique zero element represented by 0 such that $x+0 = x \quad \forall\, x \in \mathbb{F}$ additive identity

(IV) $\exists$ a unique $-x \in \mathbb{F}$ for each $x \in \mathbb{F}$ such that $x+(-x) = 0$ additive inverse

(V) Multiplication is commutative: $xy = yx \quad \forall\, x, y \in \mathbb{F}$

(VI) Multiplication is associative: $x(yz) = (xy)z \quad \forall\, x, y, z \in \mathbb{F}$

(VII) $\exists$ a unique non-zero element 1 in $\mathbb{F}$ such that $x1 = x \quad \forall\, x \in \mathbb{F}$

(VIII) To each non-zero $x \in \mathbb{F}$ $\exists\, x^{-1} \in \mathbb{F}$ such that $xx^{-1} = 1$

(IX) Multiplication distributes over addition, $x(y+z) = xy + xz \quad \forall\, x, y \in \mathbb{F}$

e.g. $A = \{1, 2, 3, 4, \ldots\}$ $\quad +$: addition, $\cdot$: multiplication $\quad$ ✗

e.g. $\mathbb{R}$, $\mathbb{C}$ ← Fields

finite field: $F_5 = \{0, 1, 2, 3, 4\}$ ✓

$+$: $(x+y) \bmod 5$ ✓

$\cdot$: $(xy) \bmod 5$ ✓

$(F_5, +, \cdot)$ is a field

$x \quad -x = 0$

(i) $(2+3) \bmod 5 = 1 \qquad (3+2) \bmod 5 = 1$

(II) $\{2+(3+4)\} \bmod 5 = 4 \qquad \{(2+3)+4\} \bmod 5 = 1$

(III) $(0+3) \bmod 5 = 3$

(IV) $(0+0) \bmod 5 = 0$, $(1+4) \bmod 5 = 0 \qquad (2+3) \bmod 5 = 0$

(V) $(2 \cdot 3) \bmod 5 = 1 \qquad (3 \cdot 2) \bmod 5 = 1$

(VI) $\{2 \cdot (3 \cdot 1)\} \bmod 5 = 1 \qquad \{1 \cdot (2 \cdot 3)\} \bmod 5 = 1$

(VII) $(1 \cdot x) \bmod 5 = x$

(VIII) $(1 \cdot 1) \bmod 5 = 1$

$(2 \cdot 3) \bmod 5 = 1$

$(4 \cdot 4) \bmod 5 = 1$

(IX) $3(2+1) = 3 \times 2 + 3 \times 1$ ✓

$\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$

Vector space: $(V, +, \cdot, \mathbb{F})$

$+$: vector addition

$\cdot$: scalar multiplication

$+$: For each $x, y \in V$ $\quad x+y \in V$

$\cdot$: For $\alpha \in \mathbb{F}$, $x \in V$ $\quad \alpha x \in V$

$x, y \in V$

$\boxed{\alpha x + \beta y \in V}$

$\alpha, \beta \in \mathbb{F}$

$(V, +, \cdot, \mathbb{F})$ is a vector space if :—

(a) $x+y = y+x \quad \forall\, x, y \in V$

(b) $x+(y+z) = (x+y)+z \quad \forall\, x, y \in V$

(c) $\exists$ a unique 0 element, $x+0 = x \quad \forall\, x \in V$

(d) $\exists$ a unique $-x$ element, $x+(-x) = 0 \quad \forall\, x \in V$

(e) $1x = x \quad \forall\, x \in V$

(f) $(\alpha\beta)x = \alpha(\beta x)$

(g) $\alpha(x+y) = \alpha x + \alpha y$

(h) $(\alpha+\beta)x = \alpha x + \beta x \in V$

e.g. $\mathbb{R}^n$ $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $x_1, x_2 \ldots x_n \in \mathbb{R}$

$+$

$\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} x+p \\ y+q \end{pmatrix}$ $\qquad \alpha \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha x \\ \alpha y \end{pmatrix}$

Q. will this be V.S?

e.g. Set of $n \times n$ matrices, Set of n-degree polynomials, C^∞ fn

$\{1, x, x^2, \ldots \ldots\}$

Linear combination: $x \in V$ then we say that x is a linear combination of $v_1, v_2 \ldots v_r \in V$ if $x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots \alpha_r v_r$, where $\alpha_1, \alpha_2 \ldots \alpha_r \in \mathbb{F}$

e.g. $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^3$ $\qquad \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1\underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{v_1} + 1\underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}_{v_2} = \frac{2}{3}\underbrace{\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}}_{b_1} + \frac{1}{3}\underbrace{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}_{b_2}$

Subspace: $V_1 \subseteq V$ and V_1 is also a vector space $\Rightarrow$ V_1 is a subspace of V

$(+, \cdot, \mathbb{F})$

e.g. $\mathbb{C}^n$ over $\mathbb{R}$ $\qquad \mathbb{R}^n \subseteq \mathbb{C}^n$ over $\mathbb{R}$

Definition (Linear dependence): Let V is a vectorspace over $\mathbb{F}$.
A $S \subseteq V$ is said to be linearly dependent if $\exists\ v_1, v_2 \cdots v_r \in S$ and $\alpha_1, \alpha_2, \cdots \alpha_r \in \mathbb{F}$, not all of which are 0 s.t.

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_r v_r = 0$$

e.g. $\mathbb{R}^3$ $\quad \underline{\underline{S}} = \left\{ \begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix} \right\} \subseteq \mathbb{R}^3$

$$1\begin{pmatrix}1\\1\\0\end{pmatrix} - 1\begin{pmatrix}1\\0\\0\end{pmatrix} - 1\begin{pmatrix}0\\1\\0\end{pmatrix} = 0$$

$\underline{\underline{S}} = \left\{ \underbrace{\begin{pmatrix}1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \begin{pmatrix}1\\1\\0\\0\end{pmatrix}}_{\subseteq \mathbb{R}^4} \right\}$

$v_1 \quad v_2 \quad v_3 \quad v_4$

$$\begin{pmatrix}1\\1\\0\\0\end{pmatrix} - \begin{pmatrix}1\\0\\0\\0\end{pmatrix} - \begin{pmatrix}0\\1\\0\\0\end{pmatrix} = 0$$

$S = \left\{ \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \right\}$ linearly indep.

$$\begin{pmatrix}1\\2\\3\\0\end{pmatrix} \notin \left\langle \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \right\rangle$$

Bases:

Let V be a vector space. A basis for V is a linearly independent set of vectors which spans V.

$\mathcal{B} \subseteq V \qquad \mathcal{B} = \{b_1, b_2, \cdots b_r\}$

$x \in V \qquad x = \alpha_1 b_1 + \alpha_2 b_2 + \cdots \alpha_r b_r \quad , \alpha_1, \alpha_2, \cdots \alpha_r \in \mathbb{F}$

e.g. $\mathbb{R}^4$ $\quad \mathcal{B} = \left\{ \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\0\\1\end{pmatrix} \right\} \subseteq \mathbb{R}^4$

$$\begin{pmatrix}x_1\\x_2\\x_3\\x_4\end{pmatrix} = x_1\begin{pmatrix}1\\0\\0\\0\end{pmatrix} + x_2\begin{pmatrix}0\\1\\0\\0\end{pmatrix} + x_3\begin{pmatrix}0\\0\\1\\0\end{pmatrix} + x_4\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

$$\mathcal{B}_1 = \left\{ \begin{pmatrix}1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\0\\1\end{pmatrix} \right\}$$

Co-ordinates: (α, β, γ)

$\alpha \to$ x-dir^n
$\beta \to$ y-dir^n
$\gamma \to$ z-dir^n

Given a vector space V and a basis $\mathcal{B} = \{b_1, b_2, \dots b_n\}$.
the co-ordinates of the vector $x \in V$ is $(\alpha_1, \alpha_2 \dots \alpha_n)$, where

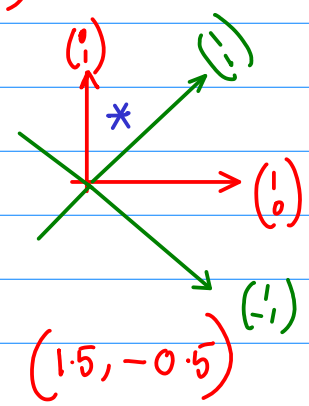
$$x = \alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_n b_n$$

e.g. $\begin{pmatrix}1\\2\end{pmatrix} \in \mathbb{R}^2$ $\quad \mathcal{B}_1 = \left\{ \underset{e_1}{\begin{pmatrix}1\\0\end{pmatrix}}, \underset{e_2}{\begin{pmatrix}0\\1\end{pmatrix}} \right\}$ $\quad 1e_1 + 2e_2$

Canonical basis $\quad (1, 2)$

$\mathcal{B}_2 = \left\{ \begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix} \right\}$ $\quad (2, 1)$

$\mathcal{B}_3 = \left\{ \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix} \right\}$ $\quad (\alpha, \beta)$ $\quad \left\{ \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix} \right\}$



$$\begin{bmatrix}1\\2\end{bmatrix} = \alpha \begin{pmatrix}1\\1\end{pmatrix} + \beta \begin{pmatrix}1\\-1\end{pmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix} \begin{bmatrix}\alpha\\\beta\end{bmatrix}$$

$$\begin{bmatrix}\alpha\\\beta\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}^{-1} \begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}1.5\\-0.5\end{bmatrix}$$

Change of basis:

$x \in \mathbb{R}^2$ $\quad \mathcal{B}_3 = \left\{ \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix} \right\}$ $\quad (1.5, -0.5)$

$\mathcal{B}_4 = \left\{ \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix} \right\}$ $\quad ? \ (\gamma_1, \gamma_2)$

$$1.5 \begin{bmatrix}1\\1\end{bmatrix} - 0.5 \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix} \leftarrow \{e_1, e_2\}$$

$$\gamma_1 \begin{bmatrix}1\\2\end{bmatrix} + \gamma_2 \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix}$$

$$1.5 \begin{bmatrix}1\\1\end{bmatrix} - 0.5 \begin{bmatrix}1\\-1\end{bmatrix} = \gamma_1 \begin{bmatrix}1\\2\end{bmatrix} + \gamma_2 \begin{bmatrix}1\\-1\end{bmatrix}$$

$$\Downarrow$$

$$\underbrace{\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}}_{P} \begin{bmatrix}1.5\\-0.5\end{bmatrix} = \underbrace{\begin{bmatrix}1 & 1\\2 & -1\end{bmatrix}}_{Q} \begin{bmatrix}\gamma_1\\\gamma_2\end{bmatrix} \Rightarrow \begin{bmatrix}\gamma_1\\\gamma_2\end{bmatrix} = Q^{-1} P \begin{bmatrix}1.5\\-0.5\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$$

$\mathcal{B}_1$ $\quad \mathcal{B}_2$

Q: Find the co-ordinates of $\begin{pmatrix}1\\2\\3\end{pmatrix}$ in $\left\{\begin{pmatrix}1\\-1\\0\end{pmatrix}, \begin{pmatrix}0\\1\\-1\end{pmatrix}, \begin{pmatrix}1\\0\\-1\end{pmatrix}\right\}$?

~~(1,2,3)~~

e_1, e_2, e_3

$S \subseteq V$ $\{b_1, b_2, \dots\}$ finite elements $\rightarrow$ finite-dimensional vector space

infinite element

($n \times n$ matrices)

$\dim(V) = \#$ basis vectors

e.g $\mathbb{R}^3$ $\dim(\mathbb{R}^3) = 3$

$\dim(V) = 4$